

DevOps Documentation

Program 10: Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring

Program-10

—> deployment.yaml

—> service.yaml

S1. Initiate Minikube allocating specific CPUs and memory minikube start --cpus=2 --memory=4096

```
1rv24mc113_varun@varun-srivastava:~/DevOpsLab/program10$ minikube start --cpus=2 memory=4096
🐳 minikube v1.37.0 on Ubuntu 24.04
🔧 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📡 Pulling base image v0.0.48 ...
🔄 Updating the running docker "minikube" container ...

💡 Docker is nearly out of disk space, which may cause deployments to fail! (87% of capacity). You can pass '--force' to skip this check.
💡 Suggestion:

Try one or more of the following to free up space on the device:

1. Run "docker system prune" to remove unused Docker data (optionally with "-a")
2. Increase the storage allocated to Docker for Desktop by clicking on:
   Docker icon > Preferences > Resources > Disk Image Size
3. Run "minikube ssh -- docker system prune" if using the Docker container runtime
🔗 Related issue: https://github.com/kubernetes/minikube/issues/9024

🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
   ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: default-storageclass, storage-provisioner, dashboard
👉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1rv24mc113_varun@varun-srivastava:~/DevOpsLab/program10$
```

S2. Enable Dashboard

minikube dashboard

```
1rv24mc113_varun@varun-srivastava:~/DevOpsLab/program10$ minikube dashboard
👉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1rv24mc113_varun@varun-srivastava:~/DevOpsLab/program10$ minikube dashboard
👉 Verifying dashboard health ...
🔧 Launching proxy ...
👉 Verifying proxy health ...
📡 Opening http://127.0.0.1:38753/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
👉 Opening in existing browser session.
```

S3. Create a Deployment.yaml file

```
! deployment.yaml x
! deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: nginx-deployment
5    labels:
6      app: nginx
7
8  spec:
9    replicas: 2
10   selector:
11     matchLabels:
12       app: nginx
13
14   template:
15     metadata:
16       labels:
17         app: nginx
18     spec:
19       containers:
20       - name: nginx
21         image: nginx:latest
22         ports:
23         - containerPort: 80
```

S4. Create a Service.yaml file

```
! service.yaml x
! service.yaml
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: nginx-service
5
6  spec:
7    selector:
8      app: nginx
9
10   ports:
11     - protocol: TCP
12       port: 80
13       targetPort: 80
14
15   type: NodePort
16
```

S5. Apply resources

kubectl apply -f deployment.yaml

kubectl apply -f service.yaml

- 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program10\$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment configured
- 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program10\$ kubectl apply -f service.yaml
service/nginx-service created
- 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program10\$ █

S6. Monitor via CLI

kubectl get pods

S7. Check nodes

kubectl get nodes

```
• 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program10$ kubectl get nodes
NAME        STATUS    ROLES    AGE   VERSION
minikube    Ready     control-plane  6d7h  v1.34.0
○ 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program10$
```

S8. Check Services

kubectl get svc

```
• 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program10$ kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1     <none>       443/TCP          5h5m
nginx-deployment    NodePort    10.105.43.174 <none>       80:32238/TCP     5h5m
nginx-service       NodePort    10.111.7.103  <none>       80:30502/TCP     91s
○ 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program10$
```

S9. Check Pod Logs

kubectl logs nginx-deployment

```
• 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program10$ kubectl logs nginx-deployment-6f9664446b-xkg4x
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/01/06 19:55:09 [notice] 1#1: using the "epoll" event method
2026/01/06 19:55:09 [notice] 1#1: nginx/1.29.4
2026/01/06 19:55:09 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/01/06 19:55:09 [notice] 1#1: OS: Linux 6.14.0-27-generic
2026/01/06 19:55:09 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/01/06 19:55:09 [notice] 1#1: start worker processes
2026/01/06 19:55:09 [notice] 1#1: start worker process 29
2026/01/06 19:55:09 [notice] 1#1: start worker process 30
2026/01/06 19:55:09 [notice] 1#1: start worker process 31
2026/01/06 19:55:09 [notice] 1#1: start worker process 32
2026/01/06 19:55:09 [notice] 1#1: start worker process 33
2026/01/06 19:55:09 [notice] 1#1: start worker process 34
2026/01/06 19:55:09 [notice] 1#1: start worker process 35
2026/01/06 19:55:09 [notice] 1#1: start worker process 36
○ 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program10$
```

S10. Describe Pods

kubectl describe pod nginx-deployment

```
1rv24mc113_varun@varun-srivastava:~/DevOpsLab/program10$ kubectl describe pod nginx-deployment
Name:          nginx-deployment-6f9664446b-xkg4x
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Tue, 06 Jan 2026 20:24:16 +0530
Labels:        app=nginx
               pod-template-hash=6f9664446b
Annotations:   <none>
Status:        Running
IP:            10.244.0.104
IPs:
  IP:          10.244.0.104
Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:  docker://eed0f804618859ed77e920aa6f61c5f857af7272edd5d107ff63534fa93b9764
    Image:          nginx:latest
    Image ID:       docker-pullable://nginx@sha256:ca871a86d45a3ec6864dc45f014b11fe626145569ef0e74deaffc95a3b15b430
    Port:          80/TCP
    Host Port:     0/TCP
    State:         Running
      Started:     Wed, 07 Jan 2026 01:25:09 +0530
    Last State:    Terminated
      Reason:      Completed
      Exit Code:   0
      Started:     Wed, 07 Jan 2026 01:06:24 +0530
      Finished:    Wed, 07 Jan 2026 01:24:42 +0530
    Ready:        True
    Restart Count: 3
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-9hssd (ro)
Conditions:
  Type                               Status
  PodReadyToStartContainers         True
```

S11. Open Minikube Dashboard

Minikube dashboard

The screenshot shows the Minikube Dashboard interface. The top navigation bar includes a search bar and a 'Relaunch to update' button. The left sidebar contains a menu with categories: Workloads, Service, and Config and Storage. Under Workloads, there are links for Cron Jobs, Daemon Sets, Deployments, Jobs, Pods, Replica Sets, Replication Controllers, and Stateful Sets. Under Service, there are links for Ingresses, Ingress Classes, and Services. Under Config and Storage, there are links for Config Maps, Persistent Volume Claims, Secrets, and Storage Classes. The main content area displays three large green circles representing the status of Deployments, Pods, and Replica Sets, each with a 'Running' label and a count. Below these, there are two tables: 'Deployments' and 'Pods'. The 'Deployments' table has columns for Name, Images, Labels, Pods, and Created. The 'Pods' table has columns for Name, Images, Labels, Node, Status, Restarts, CPU Usage (cores), Memory Usage (bytes), and Created.

Name	Images	Labels	Pods	Created
nginx-deployment	nginx:latest	app: nginx	2 / 2	6 days ago

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)	Created
program-8	node:18	app: nodejs	minikube	Running	2	-	-	52 minutes ago